

Question				Marking details	Marks available				Maths	Prac
					AO1	AO2	AO3	Total		
6	(a)	(i)		T tension in the string and mg weight of mass (gravitational force or gravity)	1			1		
		(ii)		T does not have a component tangential to the arc (1) Component of mg tangential to the arc is $mg \sin \theta$, (1) this is in the opposite direction to s (or θ) and so the negative sign (1)		3		3	1	
		(iii)		$\text{acceleration} = \frac{-mg \sin \theta}{m} = -g \sin \theta$ $= -g \theta \text{ (1) (using the approximation)}$ $\theta = \frac{s}{l} \text{ or } \sin \theta = \frac{s}{l} \text{ (1)} = -\frac{gs}{l}$		2		2	2	
		(iv)		Acceleration $\propto \theta$ (or s) measured [from a fixed position] (1) and opposite in direction (-ve) so SHM (1)			2	2		
	(b)	(i)	I	Substitution: $T = 2\pi \sqrt{\frac{l}{g}} = 2\pi \sqrt{\frac{1.2}{9.81}}$ (1) $= 2.20$ [s] (1)	1	1		2	2	
			II	$f = \frac{1}{T} = 0.45$ [5] [H]z ecf (1)		1		1	1	
		(ii)		For maximum distance along the arc $\theta_{\max} = 0.067$, also $\sin \theta_{\max} = 0.067$ As $\theta_{\max} = \sin \theta_{\max}$ (i.e. for the largest value of θ) (1) then $\sin \theta$ is equal to θ for all θ , and approximation holds for SHM. System oscillates with SHM (1)			2	2	1	
				Question 6 total	2	7	4	13	7	0